

Lovish Thukral *AI Software Developer*

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🐙 github.com/Lovish-Thukral 📁 Portfolio

Final-year Computer Science Engineering student focused on AI engineering and intelligent systems, with hands-on experience building machine learning, LLM, RAG, and backend systems. Experienced with Python, FastAPI, REST APIs, PostgreSQL, vector search, ONNX Runtime, and model deployment, with a focus on building efficient AI systems and practical intelligent applications.

📁 PROFESSIONAL EXPERIENCE

Software Developer Intern, ASWEBWORKS LLP

07/2026 – 10/2026 | Ludhiana, INDIA

- Leading backend development for Noptch, an education and career platform spanning students, universities, courses, jobs, and freelancing.
- Designed normalized PostgreSQL schemas and relational lookup structures to support new platform modules and efficient data access.
- Implemented pagination across multiple endpoints to optimize database queries and improve performance at scale.
- Solely developed the LMS backend using Node.js, Prisma ORM, and MySQL, owning API design, database integration, and business logic.
- Built exam auto-save functionality to persist student answers at 1-minute intervals while synchronizing backend state with frontend session storage.

Full Stack Developer Trainee, Vidya Corporation

06/2025 – 07/2025 | Ludhiana, INDIA

- Completed MERN-stack training through a social media application, implementing JWT authentication, REST APIs, and MongoDB schemas for posts, likes, and comments.

📁 PROJECTS

Scaler - Career & Job Automation Platform, FastAPI | PostgreSQL | Qdrant | React | RAG

- Building an AI-powered platform that aggregates job opportunities and uses RAG-based retrieval to identify and rank opportunities based on individual student profiles.
- Designing a personalized workflow that generates profile-specific resumes and cover letters based on the requirements of each opportunity.
- Developing the platform as an end-to-end system combining semantic search, vector retrieval, backend APIs, and automated application workflows.

NullBG — On-Device Background Removal, React.js | ONNX Runtime Web | WebGPU/WASM

- Built a browser-based image-processing app performing AI background removal via local ONNX inference, keeping images on-device instead of uploading to a server.
- Implemented WebGPU-accelerated inference with a WASM fallback, using a quantized ~44MB model to keep load times practical in-browser.

Sabrina AI — Offline Voice Assistant, Python | ONNX Runtime | faster-whisper | llama.cpp

- Engineered a fully offline STT → local LLM → TTS pipeline using faster-whisper, llama.cpp, and NeuTTS, eliminating cloud APIs and internet dependency
- Developed adaptive deployment logic that analyzes device resources to select the optimal model/quantization tier, achieving ~0.5-1s response latency after initial model load on consumer hardware.

Nextep AI – Intelligent Career Mentorship & Roadmap Platform,

React.js | Node.js | MongoDB | Express | Groq API | Tailwind CSS | Axios

Built an AI career mentorship platform using contextual memory and prompt orchestration to generate personalized career guidance and dynamic learning roadmaps.

🧠 SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++

Backend & Databases: FastAPI, Node.js, Express, REST APIs, PostgreSQL, MongoDB, Prisma ORM, Qdrant

AI & ML Systems: Llama Integration, ONNX Runtime, Local Inference, LLM Quantization, LLM fine-tuning, Prompt Orchestration, Vector Search

Frontend: React, Next.js, Tailwind CSS, Framer Motion

System & Tools: Linux, Git/GitHub, Postman, CI/CD, Docker, AWS

🏆 ACHIEVEMENTS

Participant - Hack-n-Win Hackathon

Engineered a full-stack AI-agent platform in a 24-hour hackathon sprint, featuring contextual memory and prompt orchestration.

Open Source Contributor | GirlScript Summer of Code (GSSOC)

Contributed to multiple high-impact repositories by optimizing codebase performance and implementing new features across diverse tech stacks.

 EDUCATION

Bachelor of Technology (B.Tech) - Computer Science Engineering, PTU, Phagwara | CGPA: 7.5+

07/2023 – Present